

$\mu \rightarrow e\gamma$

General calculation

We consider the radiative charged-lepton flavor violating decay

$$\mu(p) \rightarrow e(p-q) + \gamma(q). \quad (1)$$

The amplitude can be written as

$$\mathcal{M}(\mu \rightarrow e\gamma) = e \epsilon^{*\mu}(q) \langle e | J_\mu^{\text{em}} | \mu \rangle. \quad (2)$$

The most general matrix element compatible with Lorentz covariance is

$$\langle e | J_\mu^{\text{em}} | \mu \rangle = \bar{u}_e(p-q) [i q^\nu \sigma_{\mu\nu} (A + B\gamma_5) + \gamma_\mu (C + D\gamma_5) + q_\mu (E + F\gamma_5)] u_\mu(p). \quad (3)$$

From the Ward identity of electromagnetism,

$$\partial^\mu J_\mu^{\text{em}} = 0, \quad (4)$$

we have

$$q^\mu \langle e | J_\mu^{\text{em}} | \mu \rangle = 0. \quad (5)$$

Contracting the general decomposition with q^μ , one obtains

$$q^\mu J_\mu^{\text{em}} = i q^\mu q^\nu \sigma_{\mu\nu} (A + B\gamma_5) + \not{q} (C + D\gamma_5) + q^2 (E + F\gamma_5). \quad (6)$$

The first term vanishes because $q^\mu q^\nu$ is symmetric while $\sigma_{\mu\nu}$ is antisymmetric:

$$q^\mu q^\nu \sigma_{\mu\nu} = 0. \quad (7)$$

Since the external photon is on shell,

$$q^2 = 0, \quad (8)$$

the Ward identity reduces to

$$\bar{u}_e(p-q) \not{q} (C + D\gamma_5) u_\mu(p) = 0. \quad (9)$$

Using

$$q = p - (p-q), \quad (10)$$

and the Dirac equations

$$\not{p} u_\mu(p) = m_\mu u_\mu(p), \quad \bar{u}_e(p-q) \not{(p-q)} = m_e \bar{u}_e(p-q), \quad (11)$$

we find

$$\begin{aligned}\bar{u}_e(p-q)\not{q}(C+D\gamma_5)u_\mu(p) &= \bar{u}_e(p-q) [\not{p} - (p \not{=} q)] (C+D\gamma_5)u_\mu(p) \\ &= \bar{u}_e(p-q) [m_\mu(C-D\gamma_5) - m_e(C+D\gamma_5)] u_\mu(p).\end{aligned}\tag{12}$$

Therefore,

$$m_e(C+D\gamma_5) - m_\mu(C-D\gamma_5) = 0.\tag{13}$$

In the limit $m_e \ll m_\mu$, this implies

$$C = D = 0.\tag{14}$$

Thus the remaining amplitude is

$$\mathcal{M}(\mu \rightarrow e\gamma) = e \epsilon^{*\mu}(q) \bar{u}_e(p-q) [iq^\nu \sigma_{\mu\nu}(A+B\gamma_5)] u_\mu(p).\tag{15}$$

Using the chiral projectors

$$P_{R,L} = \frac{1 \pm \gamma_5}{2},\tag{16}$$

we can rewrite

$$A + B\gamma_5 = A_R P_R + A_L P_L,\tag{17}$$

where

$$A_R = A + B, \quad A_L = A - B.\tag{18}$$

Therefore,

$$\mathcal{M}(\mu \rightarrow e\gamma) = e \epsilon^{*\mu}(q) \bar{u}_e(p-q) i\sigma_{\mu\nu} q^\nu (A_R P_R + A_L P_L) u_\mu(p).\tag{19}$$

After summing over final spins and photon polarizations, and averaging over the initial muon spin, one obtains

$$\overline{|\mathcal{M}|^2} = m_\mu^4 e^2 (|A_R|^2 + |A_L|^2),\tag{20}$$

where we have used $m_\mu \gg m_e$.

The decay width is then

$$\Gamma(\mu \rightarrow e\gamma) = \frac{1}{16\pi m_\mu} \overline{|\mathcal{M}|^2} = \frac{m_\mu^3 e^2}{16\pi} (|A_R|^2 + |A_L|^2).\tag{21}$$

SM calculation

We now compute the contribution in the Standard Model with massive neutrinos. Working in R_ξ gauge and neglecting Goldstone bosons, the relevant diagram contains a W boson and a neutrino running in the loop.

The one-loop amplitude has the schematic form

$$\begin{aligned}
i\mathcal{M}^{(1)} = & \int \frac{d^4k}{(2\pi)^4} \bar{u}_e(p-q) \sum_k \frac{g}{\sqrt{2}} U_{ek}^* \gamma^\alpha \frac{1-\gamma_5}{2} \frac{i(\not{p}-\not{k}+m_{\nu_k})}{(p-k)^2-m_{\nu_k}^2} \frac{g}{\sqrt{2}} U_{\mu k} \gamma^\beta \frac{1-\gamma_5}{2} u_\mu(p) \\
& \times \frac{-i}{k^2-m_W^2} \left(g_{\alpha\rho} - (1-\xi) \frac{k_\alpha k_\rho}{k^2-\xi m_W^2} \right) \left[-ie \Gamma_{WW\gamma}^{\rho\sigma\mu} \right] \\
& \times \frac{-i}{(k+q)^2-m_W^2} \left(g_{\sigma\beta} - (1-\xi) \frac{(k+q)_\sigma (k+q)_\beta}{(k+q)^2-\xi m_W^2} \right) \epsilon_\mu^*(q). \quad (22)
\end{aligned}$$

The three-gauge-boson vertex is

$$\Gamma_{\rho\sigma\mu}^{WW\gamma} = (-q-k)_\sigma g_{\rho\mu} + (2k+q)_\mu g_{\rho\sigma} - (2q+k)_\rho g_{\sigma\mu}. \quad (23)$$

The fermionic propagator can be expanded as

$$\sum_k \frac{U_{\mu k} U_{ek}^*}{(p+k)^2-m_{\nu_k}^2} = \sum_k \left[\frac{U_{\mu k} U_{ek}^*}{(p+k)^2} + \frac{U_{\mu k} U_{ek}^* m_{\nu_k}^2}{(p+k)^4} + \dots \right]. \quad (24)$$

Using unitarity of the PMNS matrix,

$$UU^\dagger = \mathbb{1}, \quad (25)$$

we have

$$\sum_k U_{\mu k} U_{ek}^* = \delta_{\mu e} = 0. \quad (26)$$

Therefore the leading term vanishes, and the first surviving contribution is proportional to

$$\sum_k U_{\mu k} U_{ek}^* m_{\nu_k}^2. \quad (27)$$

Using the freedom to subtract a common mass, this can also be written as

$$\begin{aligned}
\sum_k U_{\mu k} U_{ek}^* m_{\nu_k}^2 &= -m_1^2 (U_{\mu 2} U_{e2}^* + U_{\mu 3} U_{e3}^*) + U_{\mu 2} U_{e2}^* m_2^2 + U_{\mu 3} U_{e3}^* m_3^2 \\
&= U_{\mu 2} U_{e2}^* \Delta m_{21}^2 + U_{\mu 3} U_{e3}^* \Delta m_{31}^2. \quad (28)
\end{aligned}$$

This means that differences of neutrino masses are responsible for the amount of charged lepton flavor violation generated in the SM. In the right-handed amplitude, however, the suppression is even stronger because of the chirality structure.

The loop calculation gives

$$A_R^{(1)} = \frac{g^2 e}{32\pi^2} \frac{m_\mu}{m_W^4} \sum_{k=2,3} U_{\mu k} U_{ek}^* m_{\nu_k}^2 \left[-\frac{1}{3} + \frac{3}{s-1} - \frac{1}{(s-1)^2} \left(\frac{2s \ln s}{s-1} - 1 \right) \right]. \quad (29)$$

Here s denotes the gauge parameter associated with the R_ξ gauge calculation.

Taking the limit $s \rightarrow \infty$, the loop function tends to

$$A_R^{(1)} \rightarrow A_R. \quad (30)$$

Thus the decay rate behaves parametrically as

$$\Gamma(\mu \rightarrow e\gamma) \propto \left| \sum_{k=2,3} U_{\mu k} U_{ek}^* \frac{\Delta m_{k1}^2}{m_W^2} \right|^2 \sim 10^{-44}. \quad (31)$$

EFT calculation

At the effective-field-theory level, the leading contribution to $\mu \rightarrow e\gamma$ comes from dimension-six dipole operators. The relevant terms in the effective Lagrangian are

$$\mathcal{L} \supset \frac{C_{e\mu}^{e\gamma}}{\Lambda^2} \frac{v}{\sqrt{2}} \bar{e} \sigma_{\mu\nu} P_R \mu F^{\mu\nu} + \frac{C_{\mu e}^{e\gamma}}{\Lambda^2} \frac{v}{\sqrt{2}} \bar{\mu} \sigma_{\mu\nu} P_R e F^{\mu\nu} + \text{h.c.} \quad (32)$$

Equivalently, after taking the hermitian conjugate of the second term, the amplitude can be written as

$$i\mathcal{M} = \bar{u}_e(p-q) \frac{v}{\sqrt{2}\Lambda^2} \sigma_{\mu\nu} q^\mu [C_{e\mu}^{e\gamma} P_R + C_{\mu e}^{e\gamma*} P_L] u_\mu(p) \epsilon^{*\nu}(q). \quad (33)$$

Matching this amplitude with the general parametrization,

$$\mathcal{M}(\mu \rightarrow e\gamma) = e \epsilon^{*\mu} \bar{u}_e i \sigma_{\mu\nu} q^\nu (A_R P_R + A_L P_L) u_\mu, \quad (34)$$

we identify

$$C_{e\mu}^{e\gamma} = A_R \frac{\sqrt{2}e\Lambda^2}{v}, \quad C_{\mu e}^{e\gamma*} = A_L \frac{\sqrt{2}e\Lambda^2}{v}. \quad (35)$$

Therefore,

$$A_R = \frac{v}{\sqrt{2}e\Lambda^2} C_{e\mu}^{e\gamma}, \quad A_L = \frac{v}{\sqrt{2}e\Lambda^2} C_{\mu e}^{e\gamma*}. \quad (36)$$

Substituting into the decay width,

$$\Gamma(\mu \rightarrow e\gamma) = \frac{m_\mu^3 e^2}{16\pi} (|A_R|^2 + |A_L|^2), \quad (37)$$

we obtain

$$\Gamma(\mu \rightarrow e\gamma) = \frac{m_\mu^3 v^2}{32\pi\Lambda^4} (|C_{e\mu}^{e\gamma}|^2 + |C_{\mu e}^{e\gamma}|^2). \quad (38)$$

In this EFT description, the suppression does not come from neutrino masses. Instead, it comes from the scale Λ of new physics and from the Wilson coefficients $C_{e\mu}^{e\gamma}$ and $C_{\mu e}^{e\gamma}$.